Task 14: Linux Server Hardening & Secure Configuration

Tools Used :

- Ubuntu Linux
- SSH
- UFW Firewall
- Systemctl
- APT Package Manager
- Linux File Permission Utilities (chmod, ls)
- System Log Utilities (auth.log, journalctl)
- Lynis (Security Auditing Tool)

Introduction

In this task, I performed Linux server hardening using the Ubuntu operating system. The objective of this task was to secure the Linux system by identifying security risks, reducing system vulnerabilities, and applying secure configuration settings. Linux hardening helps in protecting systems from cyber attacks such as unauthorized access, brute force attacks, malware infections, and privilege misuse.

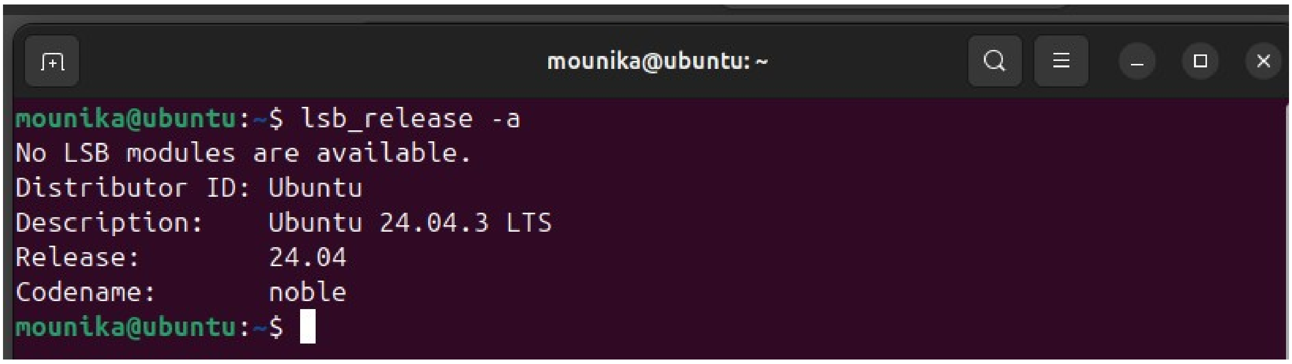
1. Review Default Linux System Settings

Initially, I reviewed the default system configuration to understand existing users, running services, and open network ports. I used system commands to check system exposure and active components.

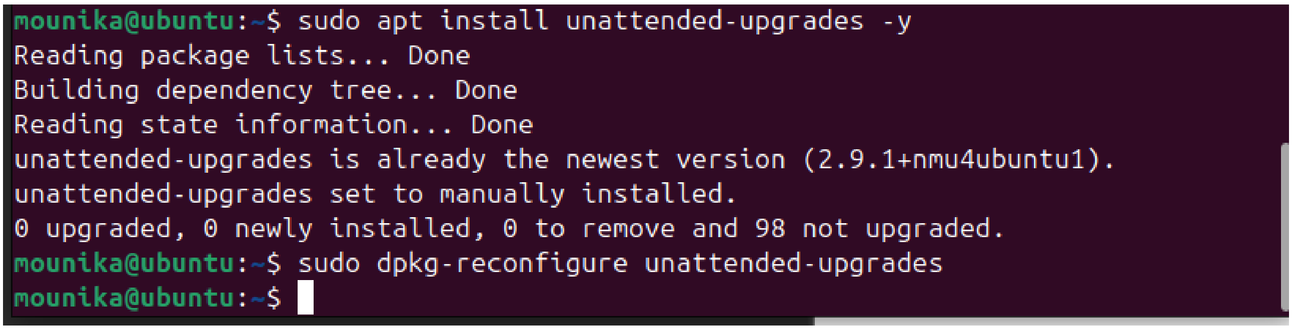
I checked system users using the `/etc/passwd` file, which shows all user accounts available in the system. I also checked currently logged-in users and verified active system services. Additionally, I checked open ports to identify which network ports were accessible.

This step helped me understand the attack surface of the system.

Screenshot: Ubuntu version



Screenshot: Auto updates enabled



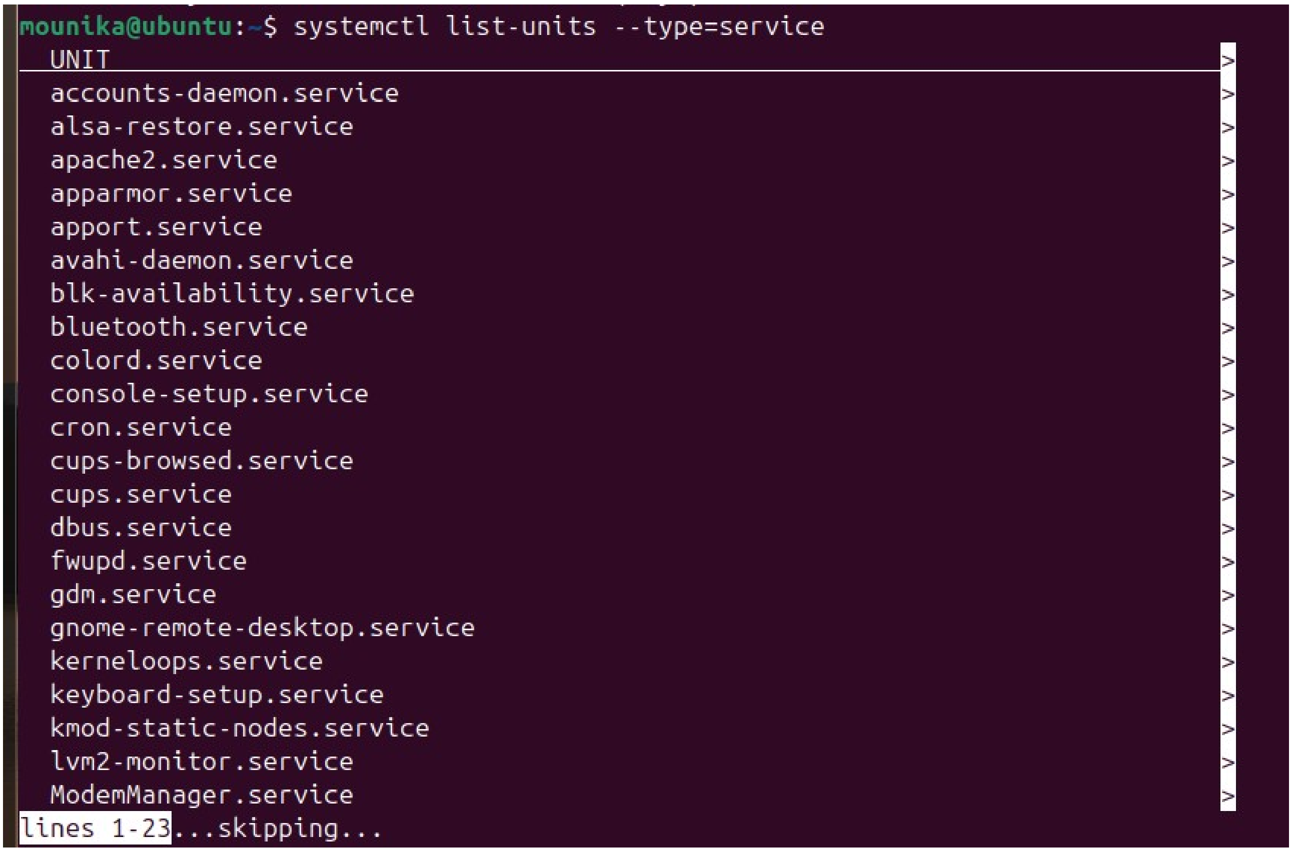
Screenshot: List of system users

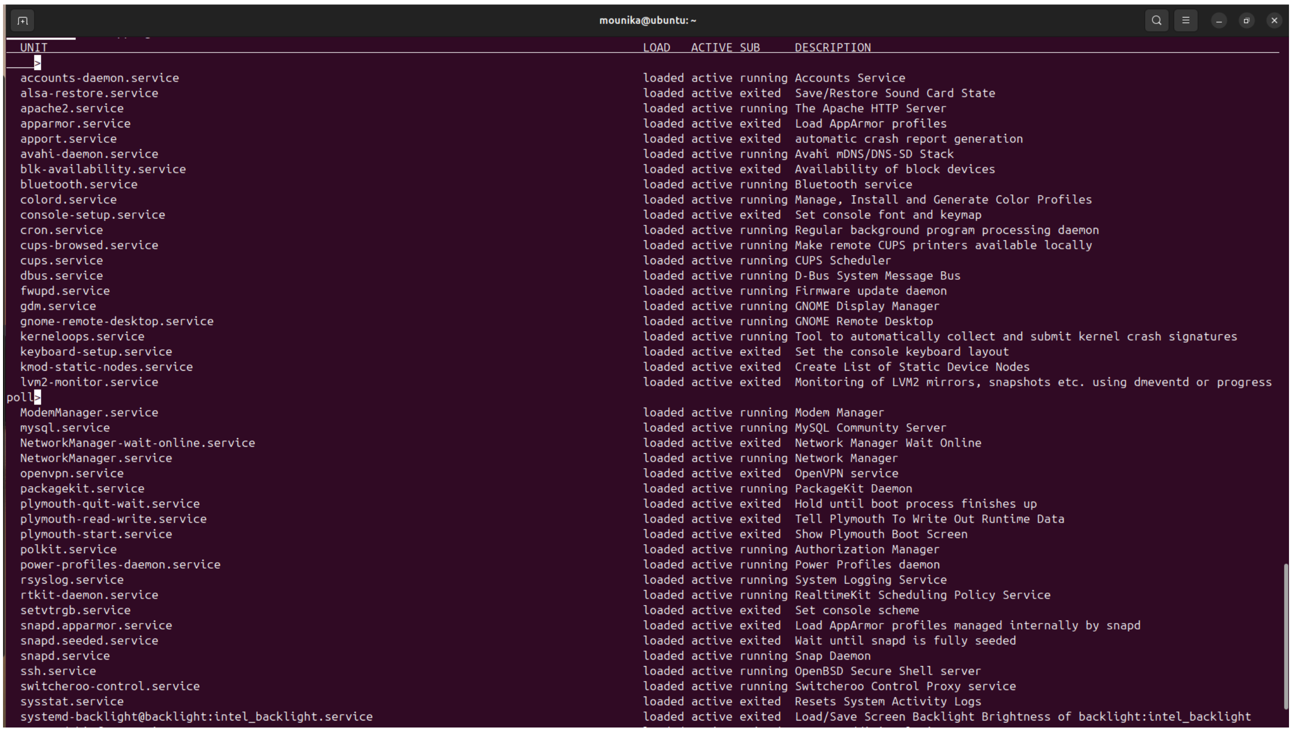
```
mounika@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
```

Screenshot: active login session

```
mounika@ubuntu:~$ who
mounika  seat0        2026-02-10 14:48 (login screen)
mounika  tty2         2026-02-10 14:48 (tty2)
mounika@ubuntu:~$
```

Screenshot: List of active services





Screenshot: Ports + LISTEN status

```
mounika@ubuntu:~$ sudo ss -tuln
Netid  State   Recv-Q  Send-Q   Local Address:Port    Peer Address:Port Process
udp    UNCONN  0       0            0.0.0.0:55863        0.0.0.0:*
udp    UNCONN  0       0         127.0.0.54:53           0.0.0.0:*
udp    UNCONN  0       0      127.0.0.53%lo:53           0.0.0.0:*
udp    UNCONN  0       0            0.0.0.0:5353         0.0.0.0:*
udp    UNCONN  0       0            0.0.0.0:54523        0.0.0.0:*
udp    UNCONN  0       0               [::]:48062           [::]:*
udp    UNCONN  0       0               [::]:5353            [::]:*
tcp    LISTEN  0       151        127.0.0.1:3306         0.0.0.0:*
tcp    LISTEN  0       4096   127.0.0.53%lo:53           0.0.0.0:*
tcp    LISTEN  0       4096      127.0.0.54:53           0.0.0.0:*
tcp    LISTEN  0       4096         0.0.0.0:22           0.0.0.0:*
tcp    LISTEN  0       4096       127.0.0.1:631          0.0.0.0:*
tcp    LISTEN  0       70         127.0.0.1:33060        0.0.0.0:*
tcp    LISTEN  0       4096           [::1]:631             [::]:*
tcp    LISTEN  0       4096            [::]:22              [::]:*
tcp    LISTEN  0       511                *:80                 *:*
mounika@ubuntu:~$
```

2. User Account and Sudo Access Management

In this step, I reviewed system users and ensured only required users had administrative access. I verified sudo privileges using group and sudo configuration checks. This step follows the principle of least privilege, which means users should have only the access required for their role.

Restricting unnecessary sudo access helps prevent privilege escalation and insider threats.

Screenshot: User in sudo group OR sudo config

```
mounika@ubuntu:~$ groups $USER
mounika : mounika adm cdrom sudo dip plugdev users lpadmin wireshark vboxusers
mounika@ubuntu:~$ sudo visudo
visudo: /etc/sudoers.tmp unchanged
mounika@ubuntu:~$
```

3. SSH Security Hardening

I secured remote login by modifying the SSH configuration file. I disabled root login and password authentication and ensured key-based authentication is enabled.

Changes applied:

- Disabled root login

- Disabled password-based login
- Enabled public key authentication

This helps prevent brute force attacks and unauthorized root access attempts.

4. System Update and Patch Management

I updated all system packages using the package manager and enabled automatic security updates. Regular system updates help fix known vulnerabilities and keep the system protected against newly discovered threats.

Automatic updates ensure the system remains secure without requiring manual intervention.

Screenshot : Change / Confirm These Values

```
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server

PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
```

```
cannot parse dst/src address.
mounika@ubuntu:~$ sudo systemctl restart ssh
mounika@ubuntu:~$ sudo sshd -T | grep -E "permitrootlogin|passwordauthentication
|pubkeyauthentication"
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
mounika@ubuntu:~$
```

5. Firewall Configuration

I configured the Uncomplicated Firewall (UFW) and allowed only required network ports. Firewall acts as a protective barrier that blocks unauthorized incoming traffic while allowing legitimate traffic.

I enabled firewall and allowed SSH port to maintain remote connectivity.

Enable Firewall

```
mounika@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
mounika@ubuntu:~$ sudo ufw allow 22
Skipping adding existing rule
Skipping adding existing rule (v6)
mounika@ubuntu:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22                         ALLOW       Anywhere
80                         ALLOW       Anywhere
443                        ALLOW       Anywhere
23                         DENY        Anywhere
Anywhere                   DENY        192.168.1.100
22 (v6)                    ALLOW       Anywhere (v6)
80 (v6)                    ALLOW       Anywhere (v6)
443 (v6)                   ALLOW       Anywhere (v6)
23 (v6)                    DENY        Anywhere (v6)

mounika@ubuntu:~$
```

6. Service Hardening

I reviewed running services and disabled unnecessary services. Each running service increases attack surface, so disabling unused services improves system security.

Screenshots : service list

```
mounika@ubuntu:~$ systemctl list-unit-files --type=service
UNIT FILE                                    STATE           PRESET
accounts-daemon.service                      enabled         enabled
alsa-restore.service                         static          -
alsa-state.service                           static          -
alsa-utils.service                           masked          enabled
anacron.service                              enabled         enabled
apache-htcacheclean.service                  disabled        enabled
apache-htcacheclean@.service                 disabled        enabled
apache2.service                              enabled         enabled
apache2@.service                             disabled        enabled
apparmor.service                             enabled         enabled
apport-autoreport.service                    static          -
apport-coredump-hook@.service                static          -
apport-forward@.service                      static          -
apport.service                               enabled         enabled
apt-daily-upgrade.service                    static          -
apt-daily.service                            static          -
apt-news.service                             static          -
autovt@.service                              alias           -
avahi-daemon.service                         enabled         enabled
blk-availability.service                     enabled         enabled
bluetooth.service                            enabled         enabled
bolt.service                                 static          -
lines 1-23...skipping...
```

Screenshot : Service disabled

```
mounika@ubuntu:~$ sudo systemctl stop bluetooth
mounika@ubuntu:~$ sudo systemctl disable bluetooth
Synchronizing state of bluetooth.service with SysV service script with /usr/lib/
systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable bluetooth
Removed "/etc/systemd/system/bluetooth.target.wants/bluetooth.service".
Removed "/etc/systemd/system/dbus-org.bluez.service".
mounika@ubuntu:~$
```

7. File Permission Security

I verified permissions of sensitive system files such as `/etc/shadow`, which stores password hashes. Ensuring correct file permissions prevents unauthorized access and protects user credentials.

Secure File Permissions

```
mounika@ubuntu:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1379 Jan 30 11:46 /etc/shadow
mounika@ubuntu:~$
```

8. Log Monitoring

I reviewed authentication logs to monitor login attempts and system activity. Log monitoring helps detect suspicious activities and possible intrusion attempts.

Check Authentication Logs

```
mounika@ubuntu:~$ sudo tail /var/log/auth.log
2026-02-10T15:23:37.876450+05:30 ubuntu dbus-daemon[1214]: [system] Rejected send message, 0 matched rules; type="error", sender=":1.73" (uid=1000 pid=2701 comm="/usr/bin/wireplumber" label="unconfined") interface="(unset)" member="(unset)" error name="org.bluez.Profile1.Error.NotImplemented" requested_reply="0" destination=":1.8" (uid=0 pid=1213 comm="/usr/libexec/bluetooth/bluetoothd" label="unconfined")
2026-02-10T15:23:37.876579+05:30 ubuntu dbus-daemon[1214]: [system] Rejected send message, 0 matched rules; type="error", sender=":1.73" (uid=1000 pid=2701 comm="/usr/bin/wireplumber" label="unconfined") interface="(unset)" member="(unset)" error name="org.bluez.Profile1.Error.NotImplemented" requested_reply="0" destination=":1.8" (uid=0 pid=1213 comm="/usr/libexec/bluetooth/bluetoothd" label="unconfined")
2026-02-10T15:23:37.889524+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-10T15:23:44.386077+05:30 ubuntu sudo:  mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/systemctl disable bluetooth
2026-02-10T15:23:44.387108+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-10T15:23:45.524507+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-10T15:25:01.232887+05:30 ubuntu CRON[8529]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-02-10T15:25:01.237199+05:30 ubuntu CRON[8529]: pam_unix(cron:session): session closed for user root
2026-02-10T15:25:39.630350+05:30 ubuntu sudo:  mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
2026-02-10T15:25:39.631049+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
mounika@ubuntu:~$
```

9. Security Audit Using Lynis

I performed a system security audit using the Lynis security auditing tool. Lynis scans the system and provides security warnings, suggestions, and hardening recommendations.

From the scan results, I observed:

- System hardening status
- Security warnings (such as system reboot recommendation)
- Security improvement suggestions

This helped me understand additional security improvements needed for the system.

Hardening Section

```
2026-02-10 15:46:35 Hardening index : [67] [#############       ]
2026-02-10 15:46:35 Hardening strength: System has been hardened, but could use additional hardening
2026-02-10 15:46:35 ====
2026-02-10 15:46:37 Tool tips: enabled
2026-02-10 15:46:37 ================================================================================
2026-02-10 15:46:37 Tests performed:     288
2026-02-10 15:46:37 Total tests:         476
2026-02-10 15:46:37 Active plugins:      2
2026-02-10 15:46:37 Total plugins:       2
2026-02-10 15:46:37 ================================================================================
2026-02-10 15:46:37 Lynis 3.1.6
2026-02-10 15:46:37 2007-2025, CISOfy - https://cisofy.com/lynis/
mounika@ubuntu:~/lynis$
```

Warnings section

```
mounika@ubuntu:~/lynis$ sudo grep -i warning /var/log/lynis.log -A 5
2026-02-10 15:46:31 Skipped test CONT-8104 (Checking Docker info for any warnings)
2026-02-10 15:46:31 Reason to skip: Prerequisites not met (ie missing tool, other type of Linux distribution)
2026-02-10 15:46:31 ====
2026-02-10 15:46:31 Skipped test CONT-8106 (Gather basic stats from Docker)
2026-02-10 15:46:31 Reason to skip: Prerequisites not met (ie missing tool, other type of Linux distribution)
2026-02-10 15:46:31 ====
mounika@ubuntu:~/lynis$
```

Hardening Index

```
mounika@ubuntu:~/lynis$ sudo grep -i "hardening index" /var/log/lynis.log
2026-02-10 15:46:35 Hardening index : [67] [#############       ]
mounika@ubuntu:~/lynis$
```

Conclusion

Linux server hardening is an essential process to secure systems from common cyber threats and unauthorized access. By applying security configurations such as restricting user privileges, securing SSH access, enabling firewall protection, updating system packages, managing services, and securing file permissions, the overall attack surface of the system was significantly reduced. Log monitoring and security auditing further helped in identifying potential risks and ensuring the system follows secure configuration standards. Proper Linux hardening ensures system stability, data protection, and improved resistance against security attacks in real-world environments.